Does a Speaking Scorer Trained on Speak & Improve Transfer to IELTS-Style Learner Speech?

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Abstract

Automated speaking assessment systems are usually tested on data from the same exam they were trained on. We ask what happens when the test format changes. We fine-tune a 3B-parameter language model on ASR transcripts from the Speak & Improve Corpus 2025 (S&I) and apply it, with no adaptation, to 45 learner recordings from the Chen (2022) English Speech Corpus, which carry IELTS Speaking bands from 5.0 to 8.0. The ranking transfers well: Spearman's ρ is 0.80 (95% CI 0.62–0.91, $n = 45$), compared with 0.71 for single S&I part responses in domain. Absolute agreement does not transfer cleanly. Predictions are compressed towards the middle of the scale, so weaker learners are over-scored by about one band and the top band is under-scored by about 0.8, and the size of the offset depends on how the S&I scale is aligned with IELTS. The test set is small and the IELTS labels have unconfirmed provenance, so we treat the results as an initial estimate.

1 Introduction

Many practice apps give learners an IELTS-style band, but few public corpora carry IELTS labels. Systems are therefore often built on data from another test, with different tasks, a different scale and different raters. Whether the resulting scores mean anything on the target test is rarely checked, because published results are almost always in domain.

The Speak & Improve Corpus 2025 [8, 9] and the Speak & Improve Challenge [14] give a large, open benchmark for scoring L2 English speech, with about 315 hours of audio scored on a CEFR-aligned scale. All challenge results are in domain. We use the corpus as a source test and ask a narrow question: if we train a text-based scorer on S&I and apply it unchanged to IELTS-style learner speech, how much of its performance survives?

We fix the whole analysis before scoring the target data (ASR settings, model, score mapping and metrics are frozen in a written protocol), score the target set once, and report ranking and absolute agreement separately. Our contributions are: (i) a pre-specified zero-shot transfer test from S&I to IELTS-style recordings; (ii) an estimate of rank agreement and absolute error, with the sensitivity of absolute error to the S&I–IELTS scale alignment; (iii) an error analysis by IELTS band that shows regression to the mean and under-scoring at the top of the range.

2 Related Work

S&I corpus and challenge. S&I contains open speaking responses from the Speak & Improve learning platform. Each submission has four marked parts, and each part receives a holistic score from 2.0 (A2) to

5.5 (C1+) in steps of 0.5 from one trained grader; the submission score is the mean of the four part scores [9]. The challenge baseline is a Whisper-small ASR cascade with a text grader [14]. In the spoken language assessment open track, the strongest systems combine speech and text models: Lin et al. [11] fuse wav2vec 2.0 with the Phi-4 multimodal LLM and report an RMSE of 0.375 (second place; first 0.364), against 0.444 for the baseline.

Text- and speech-based graders. Bannò and Matassoni [2] use wav2vec 2.0 to assess L2 spoken proficiency directly from audio, and McKnight et al. [13] grade conversational speaking tests with attention pooling over responses. Large language models are increasingly used as graders: Bannò et al. [3] show that GPT-4, used zero-shot, gives analytic scores for L2 essays that track individual proficiency components, and Ma et al. [12] fine-tune speech LLMs for holistic L2 oral proficiency, outperforming earlier baselines and generalising across speaking parts and tasks. Our scorer is text-only, which matches many practice apps and isolates what a language model learns from the words of a response.

Transfer across tests. Evaluation across corpora is less common. Bannò and Matassoni [1] use grammatical-error features learned from a written learner corpus to assess proficiency on several spoken corpora, including ICNALE, and Ma et al. [12] report generalisation across tasks within Cambridge data. Linking scales across tests is itself uncertain: IELTS is aligned to the CEFR through standard-setting studies [7, 10], and IELTS states that there is no one-to-one correspondence between bands and CEFR levels. We are not aware of earlier work that tests an S&I-trained scorer on IELTS-style speech.

3 Data

S&I (source). We use the official train/dev/eval splits. Scores are per part (P1, P3, P4, P5), range 2.0–5.5 in steps of 0.5, and correspond to A2–C1+; the criteria are modelled on Linguaskill Speaking [9]. We train on 11,858 part responses and evaluate on the eval split at part level ($n = 1200$) and at submission level ($n = 300$ test takers with all four parts). Each part has one rater. No L1 labels are available.

Chen (target). The Chen (2022) English Speech Corpus [5] contains 78 sets: 48 recordings from learners in mainland China and Hong Kong, and 30 official IELTS sample videos. We use only learner recordings. The corpus site and the classification we received list the 48 learner sets; 45 had audio available (three recordings labelled IELTS 9 are video-only and were not obtained). Each recording is one IELTS part (1, 2 or 3) of one test taker and carries one IELTS Speaking band. The 45 recordings last 3.3 hours in total (142–497 s each, median 259 s) and have bands 5.0 (9), 5.5 (12), 6.0 (3), 6.5 (6), 7.0 (6), 7.5 (3) and 8.0 (6), with equal numbers of the three parts within each band. Chen is used only as a test set: never for training, calibration or model selection. Whether the bands are the speakers' official IELTS results or ratings of these recordings is not documented; we have asked the corpus author.

How proficiency is operationalised. S&I scores are holistic CEFR-aligned grades given to one part at a time by one trained grader; the grades are not operational exam results. The Chen label is an IELTS Speaking band attached to one recording, whose provenance we could not confirm. The two quantities are related but not the same construct, and neither is a gold standard.

Scale mapping. To compare predictions with IELTS bands we first place S&I scores on the Cambridge English Scale as $\mathrm{CES} = 20s + 85$ (each half-grade owns a 10-point interval, mapped to its midpoint) and then map CES to IELTS by linear interpolation through the published Cambridge correspondence [4]: IELTS 5.0

= 154, 5.5 = 162, 6.0 = 169, 6.5 = 176, 7.0 = 185, 7.5 = 191, extrapolated linearly outside 5.0–7.5. That table is for overall B2 First and C1 Advanced scores, not speaking alone, and Cambridge claims only broad comparability. The corpus papers do not say whether an integer S&I score marks the start or the middle of a level, so we report two alternatives: integer score = start of level (+80) and middle of level (+90). Mapped metrics are therefore secondary; Spearman, which needs no mapping, is primary.

4 Method

ASR. S&I transcripts are the Whisper-small outputs shipped with the corpus (beam 5, no conditioning on previous text). Chen audio was transcribed locally with the same model (faster-whisper `small`, beam 5, `condition_on_previous_text=False`, English), so training and test input come from the same recogniser. We use ASR transcripts, not manual ones, so the scorer sees recognition errors in training.

Scorer. We fine-tune Qwen2.5-3B-Instruct [15] with QLoRA [6]: 4-bit NF4 base, LoRA rank 16 ($\alpha = 32$, dropout 0.05) on all attention and MLP projections, learning rate 2×10^{-4} with cosine schedule and 50 warm-up steps, effective batch size 16, two epochs (1,484 steps, about 4 h on two T4 GPUs). We keep the checkpoint with the lowest dev loss (0.156). The prompt contains a fixed instruction and the transcript, with no information about the part, and the target is a JSON string `{"score": s}` on the half-point grid. One model scores all parts. At test time we take the probability of each of the eight grid scores, renormalise and use the expectation $\hat{s} = \sum_k p_k s_k$. Transcripts are truncated at 900 tokens (two Chen recordings exceed this).

Evaluation protocol. All settings were frozen before Chen was scored (protocol committed 2026-10-28). The Whisper-small transcription of Chen was run after the freeze. We disclose two deviations. First, the test set was updated from 30 to 45 recordings after the freeze, before any scoring, when we received the corpus's learner/official-sample classification (our earlier split, inferred from set numbers, was wrong). Second, the planned robustness check with Whisper large-v3-turbo was dropped because of decoding loops (Section 7). Chen predictions are the raw model outputs: S&I calibrations are fitted per S&I part and have no IELTS counterpart, so no calibrated variant is reported. The unit is the recording, scored as a single response. Because each Chen recording is one IELTS part, the in-domain comparator is S&I at part level, not the submission-level score. We report Spearman's ρ (primary), Pearson's r, RMSE, bias and the share of predictions within ± 0.5 and ± 1.0 band, with 95% bootstrap intervals (2,000 resamples), and the error by reference band. For S&I, a linear calibration fitted on dev is applied to eval for the submission-level result.

5 Results

In domain. On S&I eval, single part responses give $\rho = 0.706$ (95% CI 0.674–0.735), RMSE 0.504 and 87.2% within ± 0.5 (raw, $n = 1200$). Submission-level scores (mean of four parts, calibrated) give $\rho = 0.793$ (0.745–0.833), RMSE 0.391, bias -0.069 and 83.3% within ± 0.5 ($n = 300$).

Transfer to IELTS. On Chen, raw predictions reach $\rho = 0.799$ (95% CI 0.618–0.908, $n = 45$). The interval overlaps the in-domain part-level interval, so we read this as comparable ranking and not as better transfer. The Chen set is small and its design favours rank correlation: recordings are spread evenly over bands 5.0–8.0, a wider range than the typical S&I part scores, with equal numbers of the three IELTS parts per band. On the 30 recordings with bands 5.0–6.5 alone, ρ is 0.738, and on the 15 with bands 7.0–8.0 it is 0.776; we report these post hoc and descriptively. Pearson r is 0.708.

Table 1: Chen, raw predictions, mapped to IELTS bands under three scale alignments. Spearman is 0.799 under all three. $n = 45$.

Mapping	Bias	RMSE	Within ±0.5	Within ±1.0
A: integer = start of level (+80)	+0.09	0.78	55.6%	95.6%
C: main (+85)	+0.42	0.88	48.9%	82.2%
B: integer = middle of level (+90)	+0.77	1.08	37.8%	66.7%

Table 2: Chen error by IELTS band (main mapping).

Band	n	Mean prediction	Bias
5.0	9	6.12	+1.12
5.5	12	6.54	+1.04
6.0	3	6.53	+0.53
6.5	6	6.96	+0.46
7.0	6	6.63	−0.37
7.5	3	7.14	−0.36
8.0	6	7.23	−0.77

Absolute agreement and error by band. With the main mapping the mean error is +0.42 band (RMSE 0.88; 48.9% within ±0.5, 82.2% within ±1.0). The sign and size of the bias change with the alignment (Table 1), so absolute accuracy is not a robust finding here. The error by band is more informative (Table 2, Figure 1). Mean predictions cover only about 6.1–7.2 for labels from 5.0 to 8.0: the lowest bands are over-scored by about one band (5.0: +1.12; 5.5: +1.04) and the highest band is under-scored (8.0: −0.77). The same compression appears in domain, where S&I scores of 5.0 and above are under-scored by 0.6–0.8 and scores below 3.0 are over-scored by 0.4–1.0, so it is a property of the scorer and not specific to IELTS. The under-scoring at 7.5–8.0 also fits the fact that S&I ends at C1+.

6 Discussion

A scorer trained on one test orders learners from another test about as well as it does in domain, at least in this small sample. What does not carry over is the level: the model pulls everyone towards the middle of its training range, and where that middle sits on the IELTS scale depends on an alignment that the corpus papers do not fix. In practice, a practice app built this way could rank users sensibly while telling beginners they are better than they are and advanced users that they are worse. The cheapest fix is calibration on a small labelled sample from the target test; we did not test it, since Chen is held out.

7 Limitations

- Only 45 recordings (3 of 48 learner sets were unavailable), with 3–6 per band at the top; confidence intervals are wide.
- The provenance of the Chen band labels is unconfirmed: the corpus site lists a band for each set, but we could not establish whether it is the speaker's official IELTS result or a rating of the recording.
- The learner/official-sample classification of the Chen sets comes from the corpus site and a collaborator's check of the embedded media sources, not from the corpus authors. Learners with bands 7.0–8.0 (15 recordings) come from the same hosted MP3 source as the lower bands, but if any were in

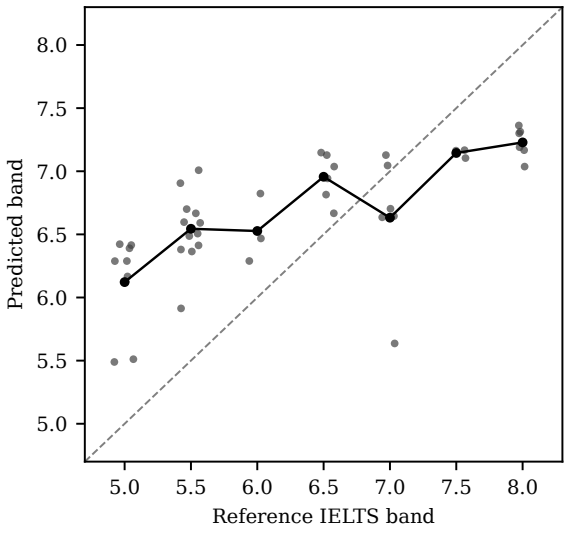


Figure 1: Predicted vs. reference IELTS band for the 45 Chen recordings (main mapping). Dashed line: identity; solid line: mean prediction per band. Predictions are compressed towards the middle of the range.

fact official sample recordings, the top-band results would change; the 5.0–6.5 subset ($n = 30$) gives $\rho = 0.738$.

- We cannot tell whether the same speaker appears in more than one set, so the 45 observations may not be fully independent.
- Each Chen recording is one IELTS part of 142–497 s; S&I responses are short per-question files, so ASR and length conditions differ. Two Chen recordings were truncated at 900 tokens.
- The Cambridge table covers overall B2 First/C1 Advanced scores, has no rows below 5.0 or above 7.5, and claims only broad comparability; our mapping extrapolates, and the choice of alignment is an assumption bounded by the A/B sensitivity rows.
- We did not run a robustness check with a larger ASR model: the Whisper large-v3-turbo transcripts we had contained decoding loops in 17 of 45 recordings, so the comparison would not have been clean.
- Text-only scorer: no pronunciation or fluency features, which the speech-based graders in related work use.

8 Conclusion

An S&I-trained text scorer ranks IELTS-style learner recordings with Spearman 0.80 ($n = 45$), similar to its in-domain part-level ranking, but its absolute scores are compressed and depend on the assumed scale alignment. Future work includes a speech-based scorer, a second test set (for example ICNALE) and confirmed IELTS labels.

Data handling and licence

S&I data were used under the corpus licence for non-commercial research. Training and scoring on ASR transcripts ran in private Kaggle notebooks; datasets were kept private and are deleted after use. No S&I data, transcripts, per-utterance outputs or trained models are released. Evaluation code will be made available. The Chen corpus is open data.

Use of AI tools

AI assistants (Claude) were used to help write and debug code and to draft and edit text. All experiments were run by the author, and all results, numbers and claims were checked by the author, who takes full responsibility for the content.

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